

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	NURUL ADRIANA BINTI KAMAL JEFRI
IC No. / Matric No.	040906-14-0680 / A23CS0258
Year / Program	2/SECPH
Section	01
Lecturer Name	DR. ROZILAWATI
Case Study	REPORT MODULE
System Name	KADA COOPERATIVE SYSTEM
Group Name	KODELAB

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SECTION A (SYSTEM'S OVERVIEW)

1.1 OVERALL DESCRIPTION

KADA is the primary agency committed to improving the living standards of farmers in the Kemubu region of Malaysia. Over the years, KADA has managed its loan and membership applications through manual processes, which only use paper form and need to insert data manually. This has proven to be inefficient, time-consuming and environmentally unfriendly. These outdated methods have resulted in issues such as record loss and delays in handling applications. As a way of addressing these issues, KADA has proposed the development of an innovative solution which is a website application named the “KADA Cooperative System” specifically to increase accessibility and achieve efficiency in the operations of membership and loan applications.

This module handles the report compilation for the Administrator and Board of Directors to aid them in decision-making. It combines both financial and operational data into well-arranged reports, including membership application total numbers, each loan type application statuses, and the overall loan summaries by month and year of the cooperative.

Table 1.1.1 Module Arrangement

Name	Matric No	Module
NABIL AFLAH BOO BINTI MOHD YOSUF BOO YONG CHONG	A23CS0252	- Authentication - Membership
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156	- Admin
LUBNA AL HAANI BINTI RADZUAN	A23CS0107	- Approval - Loan
PRAVINRAJ A/L SIVABATHI	A23CS0171	- Statement
NURUL ADRIANA BINTI KAMAL JEFRI	A23CS0258	- Report

1.2 PROJECT WEAKNESS AND IMPROVEMENT

The weakness or improvement for the report module will be represented in table as shown Table 2 below for more understanding.

Table 1.2.1 Module Weaknesses and Improvement

Module	Weakness	Improvement
Report	The charts only display the latest data, making them less useful for comparing performance over time or identifying trends.	Enhance the charting features to incorporate trend analysis and historical data, allowing users to compare data across specific months or years, such as a chart showing monthly loan approvals from the past year. For a more detailed view, enable filters by loan type or membership.
	Administrators struggle to transfer data in editable formats or perform complex analyses when only PDF output is available, limiting their flexibility.	Include the option to export data to Excel or CSV formats alongside PDF. This will enable administrators to meet diverse reporting requirements and manipulate the data, such as creating pivot tables.

1.3 PROPOSED MODULE

The current reporting module has several significant issues, particularly in its ability to display and analyze data effectively. It only presents the most recent data and lacks historical insights, which prevents Administrators and the Board of Directors (BOD) from assessing performance over time, identifying trends, or making informed, data-driven decisions. Additionally, these users struggle to perform in-depth analyses or customize reports for different use cases due to the limited export options, which are restricted to PDF format.

The existing reporting module will be replaced with a more dependable, stand-alone solution to address these gaps. The new Report Hub module aims to enhance accessibility, interaction, and flexibility by providing tools tailored to the evolving needs of BOD and KADA administrators. This change ensures that the reporting feature transforms from a static, limited data representation into a sophisticated, adaptable, and decision-oriented platform.

1.3.1 New Report Module Description

The Report Hub Module is an independent program designed to provide the Board of Directors (BOD) and administrators with advanced reporting features. It includes an annual and/or monthly report, where users can effortlessly navigate to reports by selecting a month or year.

By integrating details like both membership application and loan type application statuses, and financial summaries (such as fee capital and member funds), this report should allow users to generate detailed and customizable reports. These reports can be exported in CSV, Excel, or PDF formats and can be filtered by specific date ranges or other criteria. The module ensures easy access to essential information through its interactive features and user-friendly interface.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 CONCEPTUAL ERD

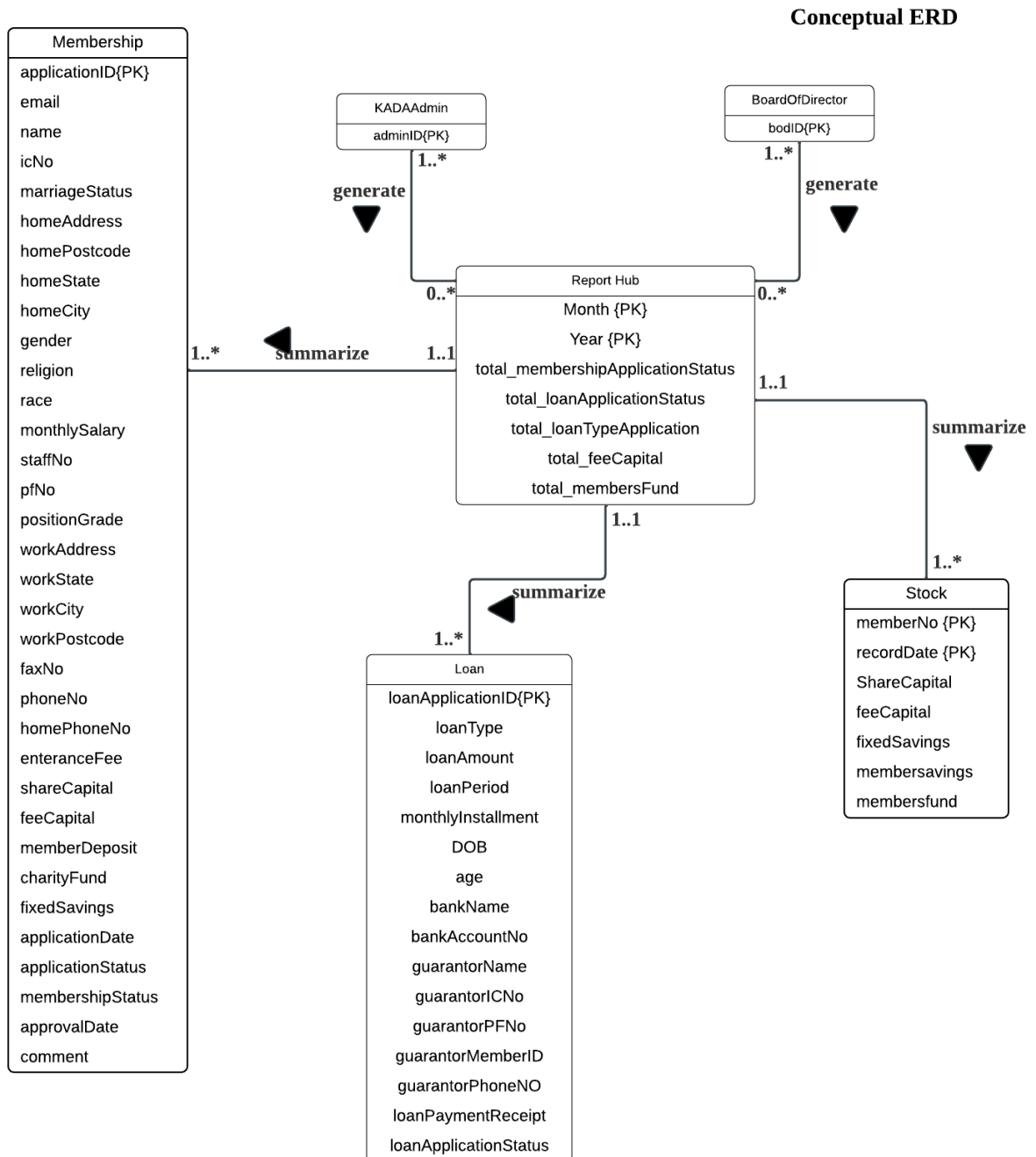


Figure 2.1.1 Conceptual ERD of Report Hub Module

2.2 LOGICAL ERD

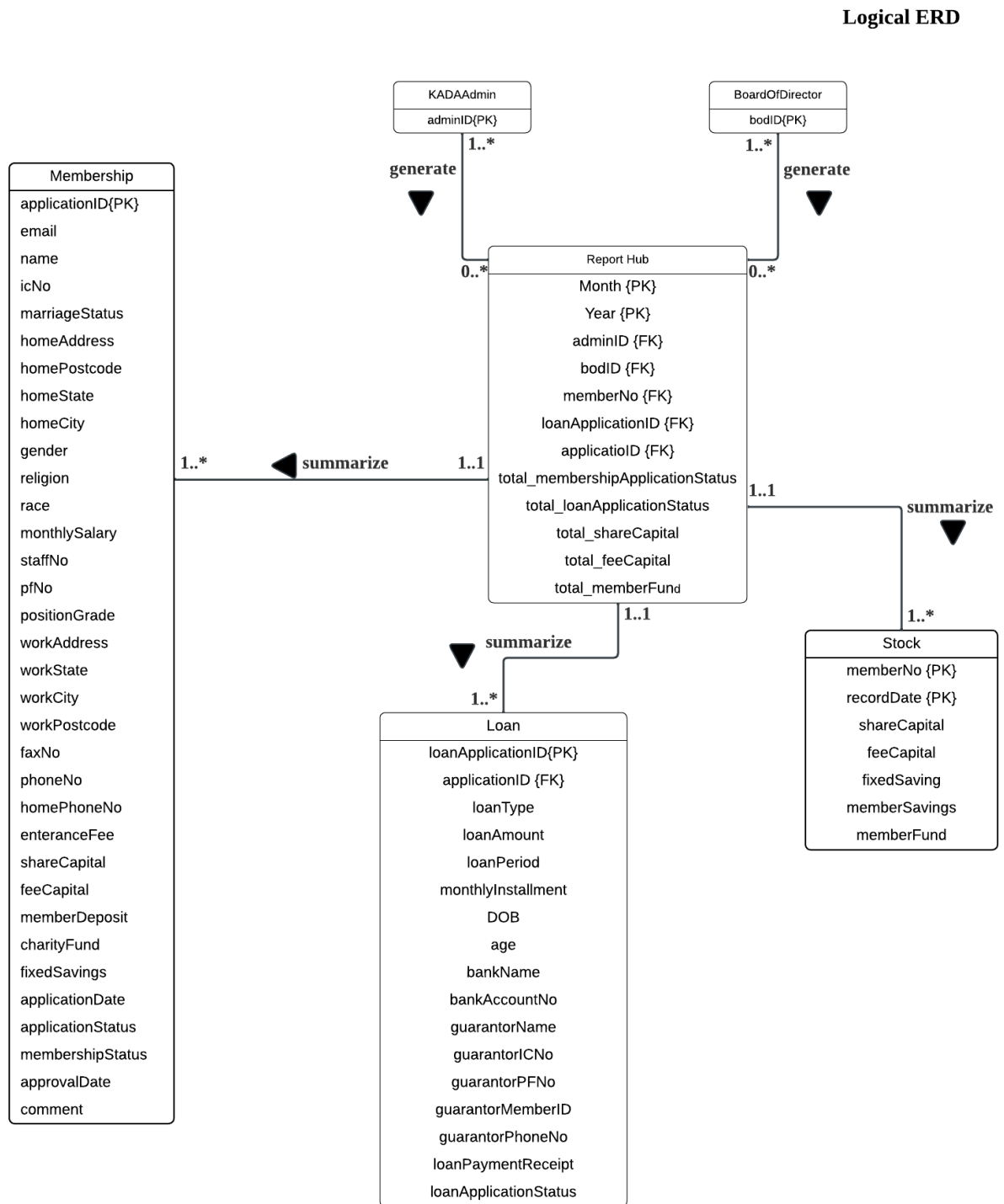


Figure 2.1.2 Logical ERD of Report Hub Module

2.3 DATA DICTIONARY FOR LOGICAL ERD

Membership

Attribute	Description	Data Type	Length	Constraints
applicationID	Unique ID for each membership application	INT	11	PRIMARY KEY, NOT NULL
email	Member's email address	VARCHAR	100	NOT NULL, UNIQUE
name	Full name of the member	VARCHAR	100	NOT NULL
icNo	Identification number of member	VARCHAR	14	UNIQUE
marriageStatus	Marital status ('Single', 'Married', 'Divorced', 'Widowed')	ENUM	-	NOT NULL
homeAddress	Residential address of the member	TEXT	-	NOT NULL
homePostcode	Postcode of home address	INT	5	NOT NULL
homeState	State of home address	VARCHAR	50	NOT NULL
homeCity	City of home address	VARCHAR	50	NOT NULL
gender	Member's gender ('Male', 'Female')	ENUM	-	NOT NULL
religion	Member's religion	VARCHAR	20	NOT NULL
race	Member's race/ethnicity	VARCHAR	20	NOT NULL
monthlySalary	Member's monthly income	DECIMAL	10, 2	NOT NULL

staffNo	Member's staff number (if applicable)	VARCHAR	50	NOT NULL
pfNo	Provident fund number	VARCHAR	50	NOT NULL
positionGrade	Employment grade level	VARCHAR	20	NOT NULL
workAddress	Office address	TEXT	-	NOT NULL
workState	State of work address	VARCHAR	50	NOT NULL
workCity	City of work address	VARCHAR	50	NOT NULL
workPostcode	Work address postcode	INT	5	NOT NULL
faxNo	Member's work phone	INT	15	-
phoneNo	Member's mobile phone	INT	15	NOT NULL, UNIQUE
homePhoneNo	Home telephone number	INT	15	-
entranceFee	Entrance fee paid by the member	DECIMAL	10, 2	NOT NULL
shareCapital	Member's share capital	DECIMAL	10, 2	NOT NULL
feeCapital	Member's fee capital	DECIMAL	10, 2	NOT NULL
memberDeposit	Member's deposit	DECIMAL	10, 2	NOT NULL
charityFund	Member's contribution to charity fund	DECIMAL	10, 2	NOT NULL
fixedSavings	Member's fixed savings	DECIMAL	10, 2	NOT NULL
applicationDate	Date the application was	DATE	-	NOT NULL

	submitted			
applicationStatus	Current status of the application ('Success', 'Pending', 'Rejected')	ENUM	-	NOT NULL
membershipStatus	Status of membership ('Active', 'Inactive')	ENUM	-	NOT NULL
approvalDate	Date the application was approved	DATE	-	-
comment	Any comments or notes on the application	TEXT	-	-

Loan

Attribute	Description	Data Type	Length	Constraints
loanApplicationID	Unique ID for each loan application	INT	11	PRIMARY KEY, NOT NULL
applicationID	Foreign key referencing Membership	INT	11	FOREIGN KEY, NOT NULL
loanType	Type of loan ('Al-Bai', 'Al-Innah', 'Baik Pulih Kenderaan', 'Road Tax dan Insurans', 'Khas', 'Karnival Musim Istimewa', 'Al-Qadrul Hassan')	ENUM	-	NOT NULL
loanAmount	Total loan amount requested	DECIMAL	10, 2	NOT NULL
loanPeriod	Loan repayment	INT	3	NOT NULL

	period in months			
monthlyInstallment	Monthly instalment amount	DECIMAL	10, 2	NOT NULL
DOB	Date of birth of the applicant	DATE	-	NOT NULL
age	Applicant's age	INT	2	NOT NULL
bankName	Name of the applicant's bank	VARCHAR	100	NOT NULL
bankAccountNo	Applicant's bank account number	VARCHAR	30	NOT NULL
guarantorName	Name of the loan	VARCHAR	100	NOT NULL
guarantorICNo	Identification card number of the generator	VARCHAR	14	NOT NULL
guarantorPFNo	Provident fund number of the guarantor	VARCHAR	20	NOT NULL
guarantorMemberID	Member ID of the guarantor	INT	11	NOT NULL
guarantorPhoneNo	Phone number of the guarantor	INT	15	NOT NULL
loanPaymentReceipt	Receipt for the loan payment	TEXT	-	
loanApplicationStatus	Status of loan application ('Approved', 'Pending', 'Rejected')	ENUM	-	NOT NULL

Stock

Attribute	Description	Data Type	Length	Constraints
memberNo	Identification number of active cooperative	INT	(11)	PRIMARY KEY, NOT

	member			NULL
recordDate	Date of recorded transaction, mainly focus on month and year	DATE	-	PRIMARY KEY, NOT NULL
applicationID	Unique ID for each membership application	INT	11	FOREIGN KEY, NOT NULL
shareCapital	The amount of money a member contribute as their share in KADA organization	DECIMAL	(12, 2)	DEFAULT 0.00
feeCapital	Total fees collected from each members	DECIMAL	(12, 2)	DEFAULT 0.00
fixedSaving	A fixed amount regularly saved by member	DECIMAL	(12, 2)	DEFAULT 0.00
memberSavings	Flexible saving contribute by the member	DECIMAL	(12, 2)	DEFAULT 0.00
memberFund	Total funds contributed or saved by all members	DECIMAL	(12, 2)	DEFAULT 0.00

Report Hub

Attribute	Description	Data Type	Length	Constraints
Month	Selected month of wanted data	INT	2	PRIMARY KEY, NOT NULL
Year	Selected year of the wanted data	INT	4	PRIMARY KEY, NOT NULL
adminID	Admin ID who generated the report	INT	11	FOREIGN KEY, NOT NULL

bodID	Board of Director ID associated with the report	INT	11	FOREIGN KEY, NOT NULL
memberNo	Identification number of active cooperative member	INT	11	DEFAULT NULL
total_membershipApplicationStatus	Summary of membership application	INT	-	DEFAULT NULL
total_loanApplicationStatus	Summary of loan applications	INT	-	DEFAULT NULL
total_loanTypeApplication	Breakdown of loan types application	INT	-	DEFAULT NULL
total_shareCapital	Total share capital reported	DECIMAL	12, 2	DEFAULT 0.00
total_feeCapital	Total fee capital reported	DECIMAL	12, 2	DEFAULT 0.00
total_memberFund	Total member's fund reported	DECIMAL	12, 2	DEFAULT 0.00
applicationID	Foreign key referencing Membership	INT	11	FOREIGN KEY
loanApplicationID	Unique ID for each loan application	INT	11	FOREIGN KEY

KADAAAdmin

Attribute	Description	Data Type	Length	Constraints
adminID	Unique identifying number for Administrator	INT	11	PRIMARY KEY, NOT NULL

BoardOfDirector

Attribute	Description	Data Type	Length	Constraints
bodID	Unique identifying number for Board of Director	INT	11	PRIMARY KEY, NOT NULL

2.4 RELATIONAL DATABASE SCHEMA

a) Membership

Attributes: applicationID (PK), email, name, icNo, marriageStatus, homeAddress, homePostcode, homeState, homeCity, gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, workAddress, workState, workCity, workPostcode, phoneNo, homePhoneNo, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, applicationDate, applicationStatus, membershipStatus, approvalDate, comment

Primary Key (PK): applicationID

b) Loan

Attributes: loanApplicationID (PK), applicationID (FK), loanType, loanAmount, loanPeriod, monthlyInstallment, DOB, age, bankName, bankAccountNo, guarantorName, guarantorICNo, guarantorPFNo, guarantorMemberID, guarantorPhoneNo, loanPaymentReceipt

Primary Key (PK): loanApplicationID

Foreign Key (FK): applicationID references Membership(applicationID)

c) Stock

Attributes: memberNo (PK), recordDate (PK), shareCapital, feeCapital, fixedSavings, memberSavings, membersFund

Primary Key (PK): Composite Key: (memberNo, recordDate)

d) Report Hub

Attributes: Month (PK), Year (PK), adminID (FK), bodID (FK), memberNo (FK), total_membershipApplicationStatus, total_loanApplicationStatus, total_loanTypeApplication, total_shareCapital, total_feeCapital, total_memberFund

Primary Key (PK): Composite Key: (Month, Year)

Foreign Key (FK): adminID references KADAAAdmin(adminID)

Foreign Key (FK): bodID references BoardOfDirector(bodID)

Foreign Key (FK): memberNo references Stock(memberNo)

e) KADAAAdmin

Attribute: adminID

Primary Key (PK): adminID

f) BoardOfDirector

Attribute: bodID

Primary Key (PK): bodID

2.5 NORMALIZATION

a) Membership

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a primary key (applicationID) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no partial dependencies; all non-prime attributes are fully dependent on the primary key, which is applicationID.
- **3NF : transitive dependencies exist.**
As shown, homePostcode depends on homeCity, and homeCity depends on homeState. Therefore, the decomposed relations in Membership is as shown below;

Membership(applicationID (PK), email, name, icNo, marriageStatus, homeAddress, homePostcode(FK), gender, religion, race, monthlySalary, staffNo, pfNo, positionGrade, entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings, applicationDate, applicationStatus, membershipStatus, approvalDate, comment)

Location => homePostcode (PK), homeCity, homeState

- **BCNF** : After 3NF normalization, this meets the requirements of BCNF.

b) Loan

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a primary key (loanApplicationID) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no partial dependencies; all non-prime attributes are fully dependent on the primary key, which is loanApplicationID.
- **3NF : transitive dependencies exist.**
As shown, age depends on DOB. Therefore, the decomposed relations in Loan is as shown below;

Loan(loanApplicationID (PK), applicationID (FK), DOB (FK), loanType, loanAmount, loanPeriod, monthlyInstallment, bankName, bankAccountNo, guarantorName, guarantorICNo, guarantorPFNo, guarantorMemberID, guarantorPhoneNo, loanPaymentReceipt)

Age => DOB (PK), age

- **BCNF** : After 3NF normalization, this meets the requirements of BCNF.

c) Stock

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a composite primary key (memberNo, recordDate) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no partial dependencies; all non-prime attributes are fully dependent on the composite primary key, which is (memberNo, recordDate).
- **3NF** : No transitive dependencies exist, so the table is already in 3NF.
- **BCNF** : This already meets the requirements of BCNF.

d) Report Hub

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a composite primary key (Month, Year) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no partial dependencies; all non-prime attributes are fully dependent on the composite primary key, which is (Month, Year).
- **3NF** : No transitive dependencies exist, so the table is already in 3NF.
- **BCNF** : This already meets the requirements of BCNF.

e) KADAAAdmin

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a primary key (adminID) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no non-prime attributes.
- **3NF** : No transitive dependencies exist, so the table is already in 3NF.
- **BCNF** : This already meets the requirements of BCNF.

f) BoardOfDirector

- **1NF** : This already meets the requirements of the first normal form (1NF) since it has a primary key (bodID) and all attributes are atomic.
- **2NF** : This already meets the requirements of the Second Normal Form (2NF) since there are no non-prime attributes.

- **3NF** : No transitive dependencies exist, so the table is already in 3NF.
- **BCNF** : This already meets the requirements of BCNF.